

BEAM and mini-BEAM

BEAM and mini-BEAM are cancer drug treatments. They are used to treat lymphoma.

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What is BEAM and mini-BEAM?

BEAM and mini-BEAM are both combinations of cancer drugs. They are used to treat [lymphoma](#).

It is best to read this information with our general information about [cancer drug treatments](#) and the [type of cancer](#) you have.

BEAM is a combination of 4 drugs:

- B – [carmustine](#) (BCNU)
- E – [etoposide](#)
- A – [cytarabine](#) (Ara-C[®])
- M – [melphalan](#).

Your cancer team will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

About BEAM

BEAM is a high-dose chemotherapy treatment. You have it during the week before a [stem cell transplant](#).

High-dose chemotherapy destroys cancer cells in your bone marrow. But it also destroys your stem cells. Stem cells are cells at the earliest stage of their development. All blood cells are made from blood stem cells.

Before you have high-dose chemotherapy, some of your stem cells are collected and stored. After the high-dose chemotherapy, the stem cells are given back to you as a drip (infusion) into a vein. The stem cells travel through the blood to your bone marrow. They then begin to make new healthy blood

cells.

About mini-BEAM

With mini-BEAM treatment, the chemotherapy drugs:

- are given in lower doses
- may be repeated after a rest period of 2 or 3 weeks.

After mini-BEAM, some people may have a stem cell transplant.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

How BEAM and mini-BEAM are given

You usually have BEAM or mini-BEAM during a stay in hospital. Some people may be able to have treatment as an outpatient. This is only suitable for some people. Your cancer team can tell you more about this.

During your course of treatment, you will meet someone from your cancer team, such as a:

- a cancer doctor
- chemotherapy nurse or specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

Before and during treatment, you will have blood tests. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your cancer drugs.

Your nurse will usually give you anti-sickness and other drugs before the cancer drugs. You will have your cancer treatment through 1 of the following:

- a [central line](#) – a fine tube that goes under the skin of the chest and into a vein close by
- a [PICC line](#) – a fine tube that is put into a vein in the arm and goes up into a vein in the chest
- an [implantable port](#) (portacath) – a disc that is put under the skin on the chest or arm and goes into a vein in the chest.

Your cancer team may also give you other medicines during your treatment. Take all these exactly as they tell you to.

Your course of chemotherapy

Your nurse or doctor will talk to you about how you have the treatment. They may give you a copy of the treatment plan.

You usually have the chemotherapy over 6 days. If you are having BEAM, the stem cell transplant is given the day after the last chemotherapy drug.

Doctors call the day you are given the stem cells Day 0. They may call the days before this day 'minus days' as they count down to the day you have your stem cells.

Day 1 (sometimes called day -6)

A nurse will give you carmustine as a drip (infusion) over 1 to 2 hours.

Days 2, 3, 4 and 5 (sometimes called days -5 to -2)

A nurse will give you:

- cytarabine twice a day as a drip over 30 minutes
- etoposide once a day as a drip over 2 to 4 hours.

Day 6 (sometimes called day -1)

A nurse will give you:

- melphalan as a drip over 30 minutes

Before and after the melphalan, you will have fluids through a drip over several hours. This is to protect your kidneys.

Stem cell transplant after BEAM

You will have the stem cells as a drip into your central or PICC line at least 24 hours after the melphalan has finished.

Mini-BEAM

The drugs are given in the same way for BEAM but the doses are lower and you will not have stem cells. After the treatment, you will have a rest period. The treatment and rest period together are called a cycle of treatment. Each cycle usually takes 21 or 28 days (3 or 4 weeks).

At the end of the 21 or 28 days, you start your next cycle of mini-BEAM. This is the same as the first cycle. Your doctor or nurse will tell you the number of cycles you are likely to have.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Side effects while treatment is being given

Some people have side effects while they are having a cancer drug or shortly after they have it.

Allergic reaction

Some people have an allergic reaction while having this treatment. Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

The drug leaks outside the vein

Sometimes cancer drugs that are given into a vein may leak outside the vein. If this happens, some drugs can damage the tissue near the vein. This is called extravasation. Extravasation is not common, but it is important that it is dealt with quickly. If you have any of the following symptoms around the vein during or after your treatment, tell your nurse straight away:

- stinging
- pain
- swelling
- if you have white skin, the area may become red
- if you have black or brown skin, the area may become darker.

Cytarabine syndrome

Cytarabine may cause some symptoms, usually about 6 to 12 hours after it has been given. Symptoms can include:

- a high temperature or chills
- a rash
- pain in the eyes, bones, tummy or chest.

This is called cytarabine syndrome. Your doctor or nurse may give you medicines such [steroids](#), to help prevent or treat it.

If you have any of these symptoms, tell your nurse. If you are having treatment as an outpatient, contact the hospital straight away on the 24-hour contact number you have been given.

Facial flushing

Carmustine may cause facial flushing while you are having it. You may suddenly feel warm, and your face may go red. This should only last a few minutes.

Most common side effects

These side effects are likely to be the most common side effects for this combination treatment. They happen to 10 or more people in every 100 people (10% or more) who have the individual drugs.

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. Your doctor may give you antibiotics and other drugs to take during this treatment. These may help reduce your risk of getting an infection.

But it is still important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

If you are having mini-BEAM, you will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

You may be given a drug called [G-CSF](#). This encourages the body to make more white blood cells. You have it as a small injection under the skin.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

-

very low in energy

- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Feeling sick

The nurses will give you anti-sickness drugs regularly. This will help prevent or control sickness during your treatment in hospital. [If you feel sick](#) or are sick (vomit), tell your nurse or doctor straight away. They can change the anti-sickness treatment. Or they might change the way you have it to something that works better for you.

If you have treatment as an outpatient, your doctor, nurse or pharmacist will prescribe anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to, even if you do not feel sick. It is easier to prevent sickness than to treat it after it has started.

Tummy pain

You may get pain in your tummy (abdomen), or have indigestion. Your doctor, nurse or pharmacist can give you advice or treatment to help. Contact the hospital straight away on the 24-hour number if your symptoms do not go away, or get worse.

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Sore mouth and throat

This treatment may cause a [sore mouth and throat](#). You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

Diarrhoea

This treatment may cause [diarrhoea](#). Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. You may also have stomach cramps. If you have a stoma, it may be more active than usual.

If you are passing loose stools 3 or more times a day and this is not normal for you, contact the hospital as soon as possible on the 24-hour number. Follow the advice they give you about:

- taking anti-diarrhoea medicines
- drinking enough fluids to keep you hydrated and to replace lost salts and minerals
- any changes to your diet that might help.

They might also ask you for a specimen of your stool to check for infection.

Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Feeling tired (fatigue)

This treatment can make you [feel very tired](#) and you will need a lot of rest. You may get tired easily for some months after treatment ends. Gentle exercise, like short walks, can help you feel less tired. The tiredness will slowly get better.

If you feel sleepy, do not drive or use machinery.

Hair loss

You usually lose all the hair from your head. You may also lose your eyelashes and eyebrows, as well as other body hair.

If you want to cover up hair loss, there are different ways you can do this. Your nurse will give you information about [coping with hair loss](#).

Remember to protect your skin from the sun. Use suncream with a sun protection factor (SPF) of at least 30 on your scalp. Or cover up with a hat or scarf.

Hair loss is almost always temporary. Your hair will usually grow back after treatment ends.

Effects on the eyes

Your eyes may become watery and feel sore and itchy. Your cancer team will give you eye drops to help prevent this. It is important to use these exactly as explained.

If you usually wear contact lenses, do not use them on the days you use eye drops or if your eyes are sore. This treatment may also make your eyes more sensitive to light and cause blurry vision. If you have pain or notice any change in your vision, always tell your doctor, nurse or pharmacist.

Effects on the lungs

This treatment can cause changes to the lungs. Tell your doctor, nurse or pharmacist if you develop:

- a cough that does not go away
- wheezing
- breathlessness.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

Effects on the liver

This treatment can affect how your liver works. You will have regular blood tests to check this.

Rarely, it can cause your skin or whites of your eyes to turn yellow (jaundice).

Headaches, dizziness or problems with balance

Carmustine can cause headaches, dizziness or difficulties with balance and walking. If you have any of these side effects, tell your doctor, nurse or pharmacist.

Low blood pressure

This treatment may cause low blood pressure. Tell your doctor or nurse if you have ever had any problems with your blood pressure. Your nurse will check it regularly during treatment.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Skin changes

Chemotherapy can affect your [skin](#). It might feel dry. Cytarabine can cause a rash, which may be itchy. It may also cause your skin to become red, blistered or peel.

If your skin feels dry, try using soap-free cleansers and unperfumed moisturising cream every day.

Rarely, this treatment can cause a serious skin reaction that needs to be treated immediately. Tell your doctor or nurse if you have any of these symptoms:

- a skin rash that is spreading
- blistering or peeling skin
- a flu-like symptoms, such as a high temperature and joint pain
- sores on your lips or in your mouth.

Always tell your doctor, nurse or pharmacist about any [skin changes](#). They can give you advice or prescribe creams or medicines to help.

During treatment and for several months afterwards, you will be more sensitive to the sun. Your skin may burn more easily than usual. You can still go out in the sun, but use a sun cream with a sun protection factor (SPF) of at least 30, and cover up with clothing and a hat. Your skin may darken. It will return to its normal colour after you finish treatment.

Effects on the heart

This treatment can affect how the heart works. You may have [tests](#) to check how well your heart is working. These may be done before, during and after treatment.

If the treatment is causing [heart problems](#), your doctor may change the type of treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Effects on the kidneys

The treatment can affect how your kidneys work. You will have blood tests before and during treatment to check this. Your nurse will ask you to drink plenty of fluids.

Your nurse will weigh you each day and also record the amount of fluid you drink and urine (pee) you pass. They will also give you extra fluids through a drip on the day you have melphalan. You may also be given a drug to make you pass more urine. This is to protect your kidneys.

If you notice blood in your urine or pass urine less often than usual, tell your nurse.

Effects on the nervous system

This high-dose chemotherapy can affect the nervous system. Tell your doctor or nurse straight away if you:

- have a headache
- feel very drowsy, confused, dizzy or unsteady
- have problems with coordination or speech
- have flickering eye movements you cannot control.

Rarely, this treatment can cause seizures (fits).

If you develop these side effects, your doctor may make changes to your treatment. It is important not to drive or operate machinery if you notice these effects.

Hand-foot (palmar-plantar) syndrome

This treatment can affect the palms of your hands and the soles of your feet. This is called palmar-plantar or hand-foot syndrome.

If you have white skin these areas may become red. If you have black or brown skin, these areas might get darker.

The skin on the palms of your hands and the soles of your feet may:

- be sore
- be painful, tingle, or swell
- peel, crack or blister.

If you have any of these symptoms, contact the hospital straight away on the 24-hour number. They can give you advice. This is especially important if you have any broken skin or if walking is difficult. They can prescribe creams and painkillers to help.

You can care for your hands and feet by:

- keeping your hands and feet cool by washing in cool water
- gently moisturising your hands and feet regularly
- wearing gloves to protect your hands and nails when working in the house or garden
- wearing loose cotton socks and avoiding tight-fitting shoes and gloves.

Tumour lysis syndrome (TLS)

Some people are at risk of developing a condition called tumour lysis syndrome (TLS). The risk is highest at the beginning of treatment. TLS happens when treatment makes large numbers of cancer cells die and break down quickly. This releases lots of waste products into the blood and can affect the kidneys and heart.

TLS can be prevented or treated. You will have regular blood tests to check for TLS. You may have:

- extra fluids through a drip
- medicines such as allopurinol tablets or [rasburicase](#) through a drip.

Drinking at least 2 litres (3½ pints) of fluid a day will also help.

Second cancer

This treatment can increase the risk of developing a second cancer years later. This is rare. The benefits of treatment usually far outweigh this risk. Your doctor, nurse or pharmacist can give you more information.

Other important information

Irradiated blood and platelets

After this treatment, if you need blood or platelet transfusions, they should be treated with radiation (irradiated) before you have them. This helps prevent a rare but serious reaction that can be caused by white blood cells in the donated blood.

Irradiated blood is not radioactive and will not harm you or anyone you come into contact with.

Some people must have any blood or platelet transfusions irradiated for the rest of their lives. If you need this, your doctor will give you more information. They will give you a medical alert card to carry in case you need blood in an emergency or at another hospital.

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Alcohol

Some preparations of this treatment contain alcohol. If having alcohol is a problem for you, tell your doctor, nurse or pharmacist. Your blood alcohol level may be above the legal limit after you have the treatment. Do not drive or operate machinery for a few hours after having this treatment, even if you feel okay.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Changes to periods

If you have periods, these may become irregular or stop while you are having this treatment. They might return after treatment, but this does not always happen. Your [menopause](#) may start sooner than it would have done. Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Sex

It is possible that small amounts of chemotherapy may be passed on through vaginal fluids or semen. If you have sex in the first few days after treatment, your cancer team will usually advise using condoms or a dental dam to protect your partner.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help